

hobbyists and anyone interested in creating interactive objects or environments.

Arduino Uno is a board based on ATmega328 microcontroller. It comprises 14 digital input/output (I/O) pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button.

It is operated with a 16MHz crystal oscillator and contains everything

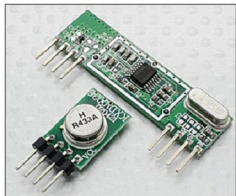


Fig. 3: 433MHz RF modules



Fig. 4: A brushless DC motor



Fig. 5: ESC module

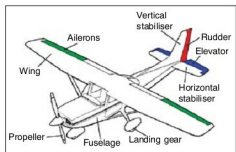


Fig. 6: Parts of an RC aircraft (courtesy: dduino.blogspot.in)

needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable, or power it with an AC-to-DC adaptor or battery to get it started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

433MHz RF modules. These are inexpensive radios that operate at 433MHz frequency (refer Fig. 3). These radios are available in a separate transmitter and receiver or a single transceiver model. The operating voltage for transmitter is from 1.5V to 12V and for the receiver is from 3V to 5V. The range of transmission is 30m to 150m depending on the voltage supply and the type of module used.

Transmitter's side. Pin 12 of board1 is connected with data pin 2 of RF transmitter (TX1). Pin A0 through A3 of board1 are connected with four 10k presets (VR1 through VR4). VR1, VR2, VR3 and VR4 are used to control BLDC (refer Fig. 4), rudder, aileron and elevator, respectively.

Receiver's side. Pin 11 of board2 is connected with data pins (6 and 7) of RF receiver (RX1). The pins 12, 10, 8 and 9 of the board are connected with the signal pin of M1 (elevator), M2 (rudder), M3 (aileron) servo motors and ESC module (refer Fig. 5), respectively. Some of the main body parts of a typical RC aircraft are shown in Fig. 6.

BLDC motor is used as the propeller system of the aircraft. The propeller

or airscrew converts rotary motion from the motor to provide propulsive force. It is the most important part of the aircraft. The propeller is mounted on the front side of the aircraft.

A rudder is used to steer the aircraft that moves through the air medium, controlling the direction in which the aircraft is pointing. It is a flat plane or sheet of material attached with hinges to the craft's stern, tail or after end.

An aileron is a hinged flight-control surface usually attached to the trailing edge of each wing of an aircraft. Ailerons are used in pairs to control the aircraft in roll, or movement around the aircraft's longitudinal axis.

Elevators are flight control surfaces, usually at the rear of an aircraft, which control the aircraft's longitudinal altitude. The position of the elevator controls whether the nose of the airplane is pointing up or down and thus moving up or down. The elevators are usually hinged to a fixed or adjustable rear surface.

Input supply wire of ESC is connected with an 11.1V battery. Three output wires of ESC are connected with three phase inputs of BLDC motor (M4).

Software

The software of this project is written in Arduino programming language. The Arduino UNO is programmed using Arduino IDE software. ATmega328 on Arduino UNO comes with a boot loader that allows you to upload a new code to it without the use of an external hardware programmer. It communicates using the STK500 protocol.

You can also bypass the boot loader and program the microcontroller through in-circuit serial programming (ICSP) header, but using boot loader programming is quick and easy. Select the correct board from 'Tools → Board' menu in Arduino IDE and burn the program (sketch) through a standard USB port in the computer.

Add library 'SoftwareServo' in the libraries folder (arduino-0022) librar-

PARTS LIST

Semiconductors:

BOARD1,	- Arduino Uno board
BOARD2	- 433MHz RF receiver
RX1	- 433MHz RF transmitter
TX1	- ESC module

Resistors (all ¼-watt, 45% carbon):

VR1-VR4	- 10-kilo-ohm preset
---------	----------------------

Miscellaneous:

CON1	- 3-pin connector
CON2	- 2-pin terminal block connector
BATT.1	- 9V PP3 battery
BATT.2	- 11.1V LiPo battery
M1-M3	- 5V servo motor
M4	- BLDC motor
	- Aircraft body